



LIST: Language-Independent Speech Token for Multilingual Speech Synthesis with Language Models

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Abstract

Recently, language model (LM)-based speech synthesis models have shown remarkable naturalness and powerful zero-shot capabilities. In this paradigm, discrete speech tokens play a critical role. Prior work has proposed using automatic speech recognition (ASR) tasks to enhance the semantic information and alignment with text in tokens. However, the commonly used byte-pair encoding (BPE) tokenizer in ASR task leads to significant differences in the text token sets of different languages, making it difficult to exploit language-shared information. This paper proposes to use the International Phonetic Alphabet (IPA) as the training target for ASR to learn language-independent speech tokens. In addition, we propose to use a timbre converter for speaker disentanglement in the speech synthesis model. Our proposed approach effectively improves the speaker similarity and expressiveness in both multilingual and cross-lingual zero-shot speech synthesis.

Index Terms: speech synthesis, multilingual, speech token, language model

1. Introduction

In recent years, neural text-to-speech (TTS) models have seen significant advancements [1, 2, 3, 4]. These models are capable of generating high-fidelity voices that are nearly indistinguishable from human speech for specific speakers. Alongside the progress in audio representation learning techniques, such as self-supervised learning (SSL) models [5, 6] and speech codec models [7, 8], several studies have incorporated continuous or discrete speech representations into speech synthesis models as intermediate representations [9, 10, 11, 12].

Recent works have demonstrated that zero-shot TTS models have the ability to synthesize speech for any given text by imitating the timbre, prosody, and style from just a few seconds of prompt audio. These models, trained on large-scale data, exhibit strong in-context learning capabilities and can be classified into two main categories. One is based on the language model (LM), which uses a speech tokenizer to extract discrete speech tokens and an autoregressive language model to predict these tokens [13, 14, 15]. Due to autoregressive sampling, LM-based models can generate diverse and prosody-consistent speech. The second category consists of non-autoregressive TTS models based on diffusion models, where flow matching is widely used and has achieved high levels of naturalness [16, 17]. Additionally, some studies have combined these two generative models to achieve even better performance [18, 19].

Obviously, speech tokens are crucial in LM-based TTS models, as they directly determine the performance of the models. While previous studies have applied discrete speech tokens to LM-based multilingual TTS models [15], they typically used unsupervised speech tokens [7, 8]. CosyVoice [19] proposed inserting a quantization layer into a multilingual ASR model to obtain discrete supervised semantic tokens, which complements lacking semantic information and alignment with text, resulting in better performance.

However, in the ASR tasks used by CosyVoice, the byte-pair encoding (BPE) tokenizer was employed to obtain text tokens as training targets. Since the BPE token sets vary significantly across different languages, it becomes challenging for the ASR model to learn shared language knowledge. As a result, the extracted speech tokens lack language-independent information, limiting the performance of multilingual LM-based TTS models that rely on these tokens, especially in cross-lingual scenarios.

To address this problem, this paper proposes using the International Phonetic Alphabet (IPA) [20] as the training target for the ASR model. IPA is a unified phonetic symbol system used to represent the pronunciations of all languages. This enables the ASR model to fully leverage the consistency of pronunciation across languages and learn language-independent semantic information. To obtain a language-independent speech tokenizer, we insert a vector quantization layer into the encoder of ASR model with IPA as training targets. Then, we extract language-independent speech tokens from the proposed speech tokenizer and integrate them into an LM-based multilingual TTS model, which autoregressively predicts the speech tokens using a language model and employs a flow matching model to generate mel spectrogram from these tokens, conditioning on speaker embedding and mel spectrogram.

Furthermore, we propose an approach to achieve speaker disentanglement during the training of the flow matching model. This method uses a timbre converter to modify the timbre attribute of the speech before extracting speech tokens and then predicts the mel spectrogram extracted from the original speech. In this way, the flow matching model is forced to rely on speaker information from the provided speaker embedding rather than the speech tokens, resulting in higher speaker similarity and improved naturalness.

Our proposed TTS model is trained on a multilingual dataset consisting of six languages. Experimental results demonstrate that the proposed method effectively improves the speaker similarity and expressiveness of the synthesized speech in both multilingual and cross-lingual zero-shot speech synthesis tasks. Audio demos, codes and models are available at <https://ryuclc.github.io/LIST>.

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2. Proposed Method

2.1. Language-Independent Speech Tokenizer

The Automatic Speech Recognition (ASR) task involves transcribing speech into text. Previous studies have explored leveraging ASR tasks to enrich the semantic information within speech representations and improve their alignment with text. For instance, Liu et al. [21] utilized the continuous hidden representations of the ASR model as intermediate representations for TTS models. In contrast, CosyVoice [19] introduced the idea of inserting a quantization layer into the ASR encoder to obtain discrete speech tokens.

As shown in Figure 1, our proposed language-independent (LI) speech tokenizer is trained using an ASR task. It first down-samples the waveform to 50 Hz representations using the same CNN encoder as HuBERT [6]. The Token encoder then extracts contextual features from these representations, consisting of the first six Transformer layers of the ASR encoder. Following this, a vector quantizer is inserted after the Token encoder to obtain discrete speech tokens, following the approach of CosyVoice. We employ a vector quantizer with a Euclidean codebook, updated using exponential moving average (EMA). After quantization, discrete speech tokens are obtained. Before feeding them into the subsequent networks, additional position embeddings are added. The Aggregation encoder consists of the remaining layers of the ASR encoder, and its output is fed to the ASR decoder to predict targets in a teacher-forcing mode during training with cross entropy (CE) loss.

In CosyVoice, the Byte-Pair Encoding (BPE) tokenizer was used to generate text tokens as training targets for multilingual ASR. However, the set of these text tokens varies significantly across different languages, which makes it challenging for the ASR model to learn language-shared semantic information. This variation negatively impacts the performance of speech tokens and, consequently, the synthesis quality when applied to the downstream multilingual TTS task.

This paper proposes converting all text transcriptions into the International Phonetic Alphabet (IPA) and using these IPA transcriptions as the training targets for the multilingual ASR model. IPA is a unified phonetic symbol system that represents the pronunciations of all languages. In this way, we can effectively leverage the consistent pronunciation features across different languages, enabling the model to learn a language-independent speech token.

Note that the encoder of our ASR model is initialized using a pretrained multilingual HuBERT model to enhance performance, given the limited computational and data resources. Additionally, the original HuBERT models operate at a frame rate of 50 Hz. Following previous work [22], we reduce this to 25 Hz without introducing extra parameters. Since HuBERT uses convolutional position embeddings, we modify the stride of the position convolution before the first encoder layer to 2 and apply average pooling on the 50 Hz representations, resulting in a 25 Hz speech token.

2.2. Language Model-Based TTS Model with Language-Independent Speech Token

As shown in Figure 2, our TTS model architecture follows the design of CosyVoice [19]. A language model is used to predict speech tokens from text. These tokens are then processed by a flow matching model to predict the mel-spectrogram, conditioned on both the speaker embedding and the reference mel-spectrogram. Finally, a vocoder is employed to reconstruct the

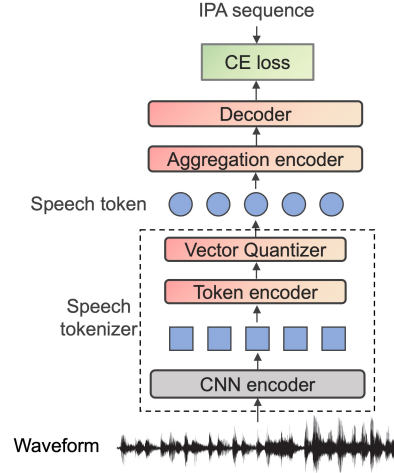


Figure 1: Proposed language-independent speech tokenizer trained by a ASR task with IPA targets. The gray module is frozen during training. The dashed-line box represents the speech tokenizer.

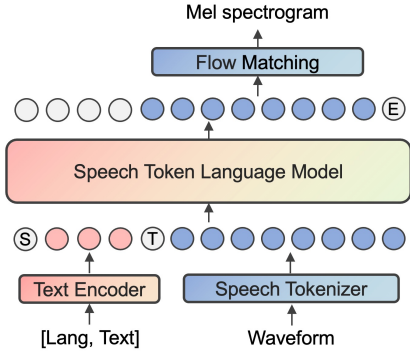


Figure 2: The LM-based TTS model used in this paper. S, E and T denote the “start of sequence”, “start of sequence” and “turn of speech” tokens respectively.

waveform.

In CosyVoice, a speaker embedding is fed into the language model. However, during our preliminary experiments, we found that this approach leads to a serious information leakage problem, particularly in cross-lingual scenarios, as also noted in CosyVoice2 [23]. As a result, we remove the speaker embedding from the language model, as shown in Figure 2. Additionally, to improve language control, we randomly prepend a language prefix to the text during language model training with a probability of 0.5.

Furthermore, to improve speaker similarity and the naturalness of synthetic speech, we propose a speaker disentanglement method during the flow matching training process. As shown in Figure 3, we use a timbre converter to modify the timbre attributes of the speech before extracting speech tokens, and then predict the mel-spectrogram based on the unaltered speech. This approach forces the flow-matching model to rely on speaker information from the provided speaker embedding and reference mel-spectrogram, rather than from the speech tokens, to achieve speaker disentanglement.

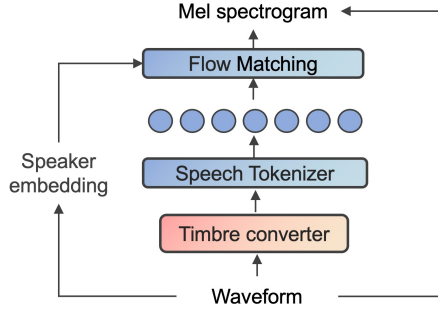


Figure 3: Flow matching with timbre converter.

3. Experiments

3.1. Datasets

3.1.1. ASR data

To train the ASR model described above, we used data from 40 languages in the Common Voice dataset [24], including Arabic (ar), Belarusian (be), Bashkir (ba), Bulgarian (bg), Catalan (ca), Chinese (zh), Czech (cs), Welsh (cy), Dutch (nl), English (en), Estonian (et), Basque (eu), Persian (fa), Finnish (fi), French (fr), Georgian (ka), German (de), Hungarian (hu), Indonesian (id), Italian (it), Latvian (lv), Lithuanian (lt), Marathi (mr), Polish (pl), Portuguese (pt), Romanian (ro), Russian (ru), Slovak (sk), Slovenian (sl), Spanish (es), Swahili (sw), Swedish (sv), Tamil (ta), Turkish (tr), Ukrainian (uk), Urdu (ur), Uzbek (uz), Uyghur (ug), Chuvash (cv), Tatar (tt). The duration of each language ranged from 50 to 200 hours, for a total of 4879 hours. Each utterance had a corresponding text transcription.

3.1.2. TTS data

To evaluate the performance of the LM-based TTS model using our proposed LI speech tokens, we used a multilingual dataset consisting of six languages. The Chinese data was obtained from Wenetspeech4TTS [25], with a duration of 945 hours, while the English data came from LibriTTS [26], with a duration of 585 hours. For the other languages, we used German, French, Japanese, and Korean data from the Emilia dataset [27], with durations of 1,590, 1,381, 1,715, and 217 hours, respectively.

To demonstrate the language independence of the speech tokens, Japanese and Korean in the TTS data were considered unseen languages in the ASR task.

3.2. Experimental setup

3.2.1. ASR

To generate the IPA training targets, we employed the Phonemizer tool [28] to convert the transcriptions of the ASR data into IPA sequences. However, this method was not applicable to Chinese due to its low accuracy. Instead, we first utilized pypinyin¹ to convert Chinese characters into pinyin, and then employed a pinyin-to-ipa tool² to convert these pinyin into IPA with tones. As a result, the size of the IPA set was 417.

As previously mentioned, the encoder of our ASR model was initialized with a pretrained multilingual HuBERT model named mHuBERT-147 [29]. We implemented an approach to

compress it to a token rate of 25 Hz without adding any extra parameters. Upon completion of the training process, the Character Error Rate (CER) of the ASR model on the validation sets of 40 languages was measured at 3.57%.

To verify the effectiveness of our proposed language-independent speech tokens, we also trained another ASR model with the same architecture and data, but using the BPE tokens from CosyVoice instead of IPA. The size of BPE token set was 43,319, resulting in a CER of 10.24% on the validation sets of 40 languages.

3.2.2. TTS

We adopted the identical model configurations as those of the original CosyVoice model. Besides, to prevent information leakage, we eliminated the speaker embedding from the Language Model (LM). For the timbre converter, we used OpenVoice [30]. Regarding the speaker embedding, we utilized the campluss [31] model to extract it from speech waveforms.

In our experiments, we compared the following three models:

- **Token**: The TTS model trained using the speech tokens extracted from our ASR model, with BPE tokens as the targets.
- **LIST**: Our proposed model, which uses the speech tokens extracted from our ASR model with IPA as the targets.
- **LIST-tc**: Our proposed model, enhanced with a timbre converter for speaker disentanglement.

3.3. Experimental results

3.3.1. Evaluation sets

To evaluate the performance of the multilingual TTS models, we prepared both an objective set and a subjective set for each language. The objective set is used to assess various objective metrics of the model. For Chinese and English, we directly used the open-source benchmarks from SEED [18], which contain 2,000 and 1,000 samples, respectively. For the other four languages, we extracted 1,000 samples each from the test sets of the Common Voice dataset. The subjective set is used for listening tests, as it includes high expressiveness and diverse styles. Each language contains approximately 100 samples. Additionally, to evaluate the model’s performance in cross-lingual scenarios, we constructed cross-lingual test sets by using the test sets from different languages.

3.3.2. Objective evaluation

We used two objective metrics: 1) character error rate (CER), which is used to evaluate intelligibility. For languages other than Chinese, we used whisper-large-v3 [32] to transcribe the generated speech. For Chinese, we used Paraformer-zh [33]. The CER is then calculated based on the transcriptions. 2) speaker similarity (SIM), which is obtained by calculating the cosine similarity between the speaker embeddings of the generated speech and the reference speech. These embeddings are extracted using a speaker verification model fine-tuned on WavLM³.

The objective evaluation results of zero-shot and cross-lingual zero-shot TTS are shown in Table 1 and Table 2, respectively. Firstly, in terms of the CER metric, both LIST and LIST-tc are generally superior to Token. This indicates that

¹<https://github.com/mozillazg/python-pinyin>

²<https://github.com/stefantaubert/pinyin-to-ipa>

³https://github.com/microsoft/UniSpeech/tree/main/downstreams/speaker_verification

Table 1: Objective evaluation results of zero-shot TTS with different models on six languages, including character error rates (CER) (%) and speaker similarity (SIM). Avg denotes the average values of six languages.

Model	zh		en		ja		ko		de		fr		Avg	
	CER↓	SIM↑	CER	SIM	CER	SIM	CER	SIM	CER	SIM	CER	SIM	CER	SIM
Human	1.14	0.755	1.21	0.737	8.57	0.708	7.41	0.716	2.45	0.762	5.13	0.740	4.32	0.736
Token	5.06	0.596	3.79	0.551	12.03	0.582	12.53	0.609	6.60	0.576	8.34	0.579	8.06	0.582
LIST	3.89	0.615	4.06	0.563	9.73	0.588	10.52	0.621	3.75	0.600	8.26	0.584	6.70	0.595
LIST-tc	4.27	0.622	4.12	0.567	9.95	0.593	10.57	0.626	3.76	0.603	8.36	0.587	6.84	0.600

Table 2: Objective evaluation results of cross-lingual zero-shot TTS with different models on six languages, including character error rates (CER) (%) and speaker similarity (SIM). Avg denotes the average values of six languages.

Model	zh		en		ja		ko		de		fr		Avg	
	CER↓	SIM↑	CER	SIM	CER	SIM	CER	SIM	CER	SIM	CER	SIM	CER	SIM
Token	3.49	0.468	3.35	0.372	15.63	0.441	9.59	0.449	3.36	0.412	7.84	0.399	7.21	0.424
LIST	3.52	0.465	3.00	0.388	16.47	0.455	8.26	0.462	2.74	0.421	7.47	0.412	6.91	0.434
LIST-tc	3.76	0.474	3.10	0.394	16.44	0.464	8.89	0.466	2.87	0.428	7.81	0.415	7.15	0.440

Table 3: MOS results of zero-shot TTS with different models on six languages with 95% confidence intervals. Avg denotes the average MOS of six languages.

Model	zh	en	ja	ko	de	fr	Avg
Token	4.27 ± 0.07	3.91 ± 0.10	4.09 ± 0.09	3.99 ± 0.10	4.08 ± 0.09	4.03 ± 0.08	4.06 ± 0.09
LIST	4.39 ± 0.06	4.06 ± 0.08	4.28 ± 0.09	4.15 ± 0.08	4.25 ± 0.09	4.19 ± 0.08	4.22 ± 0.08
LIST-tc	4.33 ± 0.07	4.13 ± 0.07	4.31 ± 0.08	4.21 ± 0.08	4.35 ± 0.08	4.23 ± 0.08	4.26 ± 0.08

Table 4: MOS results of cross-lingual zero-shot TTS with different models on six languages with 95% confidence intervals. Avg denotes the average MOS of six languages.

Model	zh	en	ja	ko	de	fr	Avg
Token	4.20 ± 0.07	4.13 ± 0.07	4.10 ± 0.07	4.12 ± 0.09	4.07 ± 0.08	4.11 ± 0.08	4.12 ± 0.08
LIST	4.31 ± 0.06	4.28 ± 0.07	4.28 ± 0.08	4.24 ± 0.09	4.19 ± 0.09	4.22 ± 0.09	4.25 ± 0.08
LIST-tc	4.36 ± 0.07	4.42 ± 0.06	4.35 ± 0.06	4.34 ± 0.08	4.32 ± 0.08	4.27 ± 0.07	4.34 ± 0.07

the speech tokens extracted from the ASR model with IPA as the targets capture more semantic information. However, the CER of LIST-tc is slightly higher than that of LIST, which may be due to the timbre converter affecting the content information. Regarding the SIM metric, LIST shows a significant improvement over Token. Meanwhile, LIST-tc further enhances speaker similarity compared to LIST. These experimental results demonstrate the superiority of the language-independent speech tokens we proposed.

3.3.3. Subjective evaluation

Subjective listening tests were conducted to evaluate the mean opinion score (MOS) of the synthetic speech. For each language, we randomly selected 20 samples from the subjective evaluation set for zero-shot synthesis. Then, we conducted subjective listening tests on the crowdsourcing platform Amazon Mechanical Turk⁴. Approximately 15 native speakers participated in each language group. These native speakers first listened to a prompt reference audio, and then rated the synthesized speech in terms of speech quality, expressiveness, and speaker similarity to the prompt audio, on a scale from 1 to 5, with intervals of 0.5.

The MOS results of zero-shot and cross-lingual zero-shot

TTS are shown in Table 3 and Table 4, respectively. The p-value of paired t-test was employed to measure the significance of the difference between two models. It is evident that LIST significantly outperforms Token in all cases ($p < 0.05$). This indicates that our proposed method leverages the unified pronunciation characteristics of different languages through IPA, facilitating speech tokens to contain more language-independent semantic information, thereby improving the performance. Meanwhile, LIST-tc also generally outperforms Token significantly ($p < 0.05$), except in zero-shot TTS for Chinese (zh) ($p > 0.05$). Compared to LIST, LIST-tc generally shows a slight but not significant improvement ($p > 0.05$).

4. Conclusion

In this paper, we proposed a novel approach to obtain language-independent speech tokens for multilingual TTS. We built an ASR model using IPA as training targets and inserted a vector quantizer into the encoder to generate discrete speech tokens. The use of IPA enables the ASR model to learn useful language-independent semantic information. Besides, we introduced a speaker decoupling method during the training process of the TTS model. This approach employs a timbre converter to modify the speech before extracting speech tokens for flow matching, resulting in improved speaker similarity and voice quality.

⁴<https://www.mturk.com/>

5. References

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